DO NOW — RECALL

Dijkstra: shortest paths II

Name: _____ Class: _____ Date: _____

1. What vertex does Dijkstra settle next?

2. What does a weighted edge record?

3. What does a predecessor in a path mean?

4. Distance to A=3, edge A–B=4, current distance B=10. Update B.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

The unsettled vertex with the smallest current tentative distance.

2. Model answer

The cost of travelling between two connected vertices.

3. Model answer

The previous vertex on that route.

4. Model answer

Candidate 3+4=7, so update B to 7 and predecessor(B)=A.

Vocabulary — know these terms

predecessor table — For each vertex, stores the prior vertex on its best-known path.

relaxation — Replace a distance when a cheaper route is discovered.

settled — Shortest distance is final for a vertex under non-negative edge weights.

infinity — Marker for an as-yet unreachable vertex.

route reconstruction — Follow predecessor links backwards then reverse.

NEW LEARNING

Read, annotate and explain

Track improvements

Relax each edge by comparing the candidate total with the current best-known distance.

Remember predecessors

When a distance improves, store the vertex from which the improvement came.

Finish with the best distance

Here E reaches distance 7 after A, C, B and D have settled.

Reconstruct backwards

Follow E←D←B←C←A, then reverse it to get the forward route.

Handle unreachable vertices

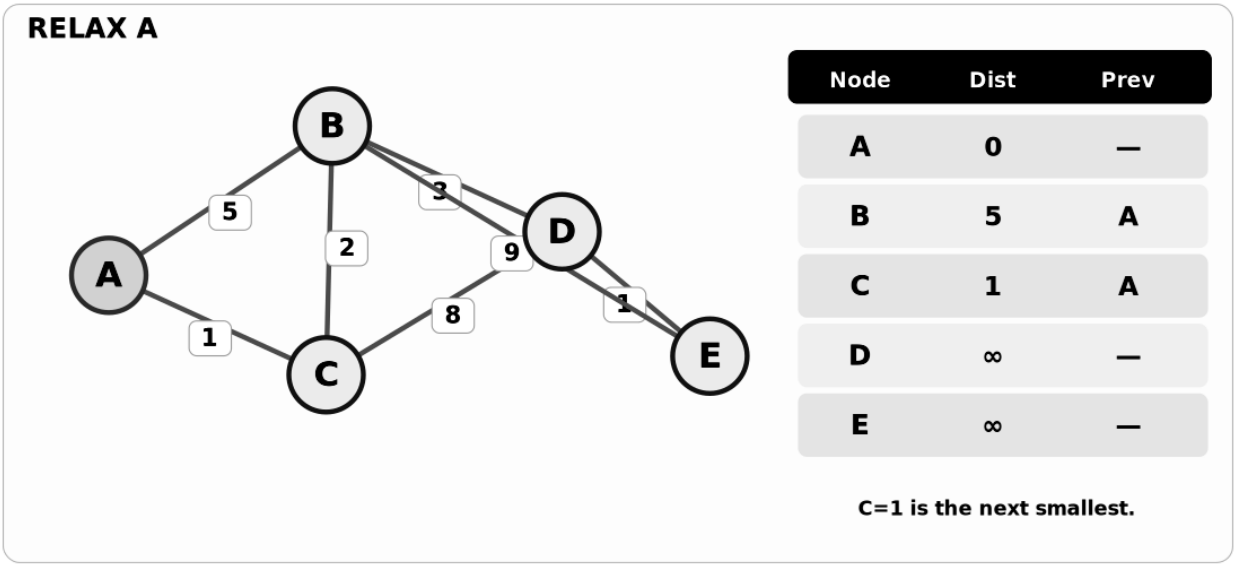
A vertex remaining at infinity cannot be reached from this start along the graph's edges.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

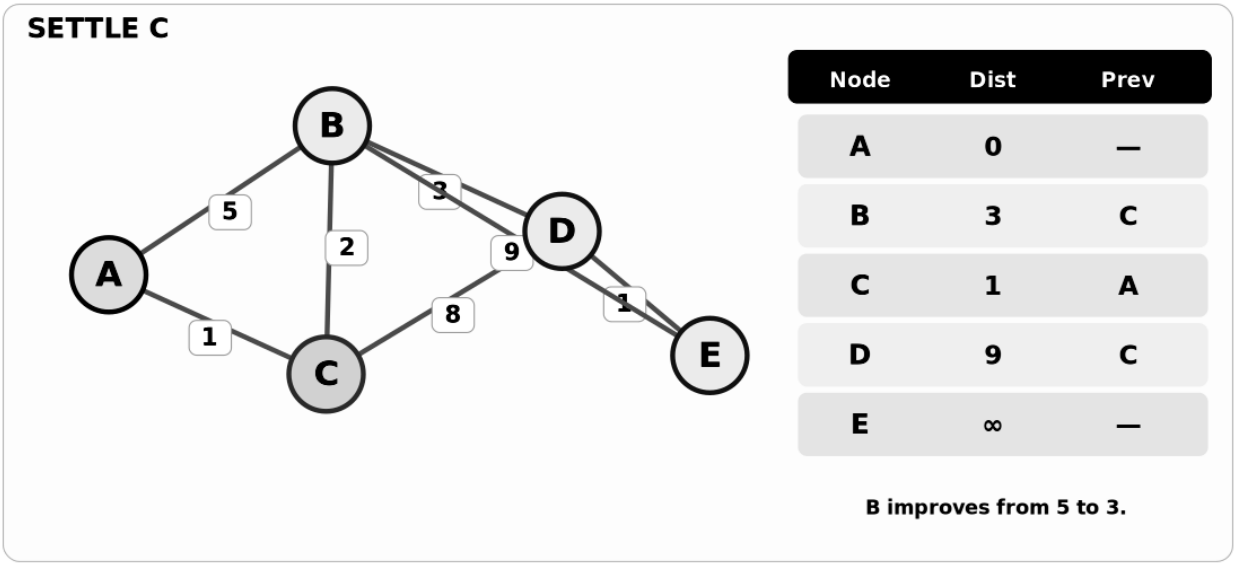
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



RELAX A Dijkstra: C=1 is the next smallest.

New Learning

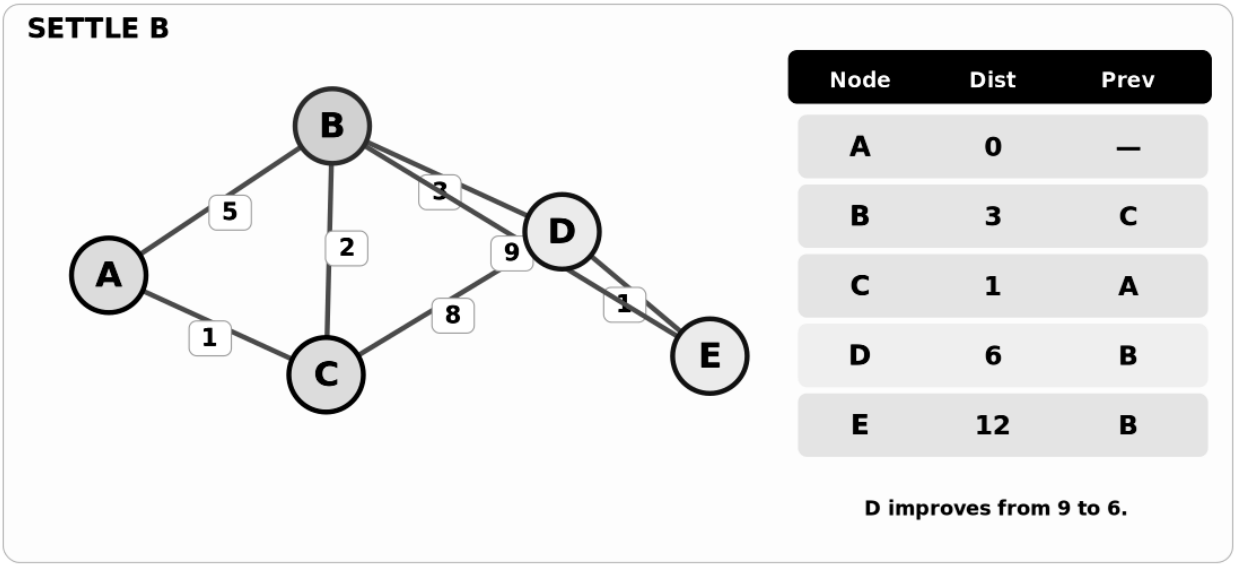


SETTLE C Dijkstra: B improves from 5 to 3.

New Learning · original lesson diagrams

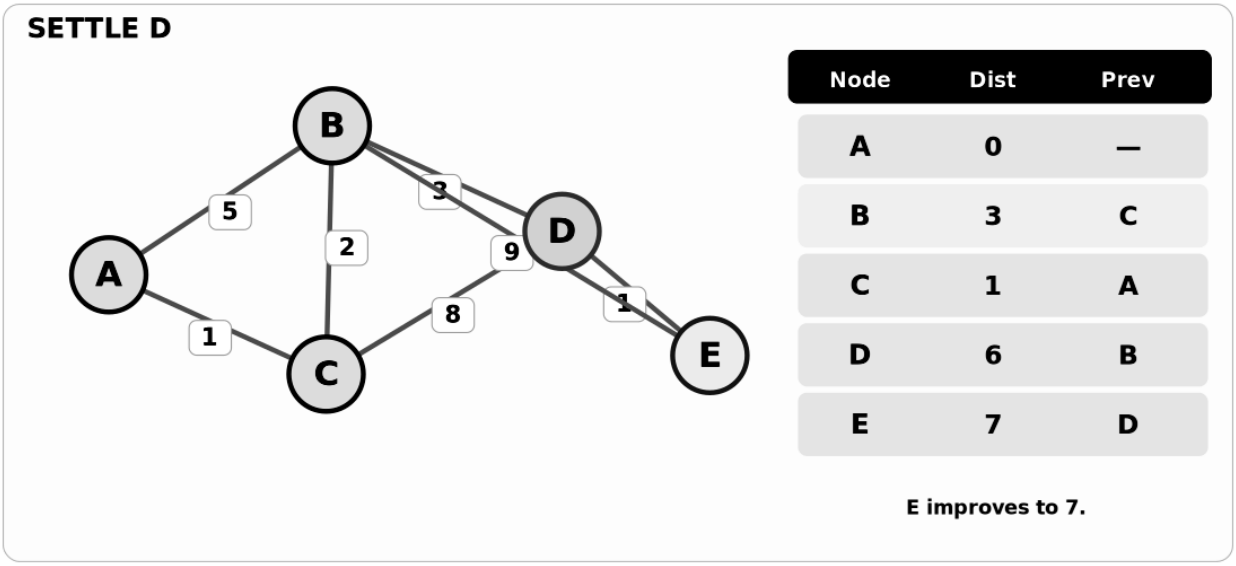
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



SETTLE B Dijkstra: D improves from 9 to 6.

New Learning

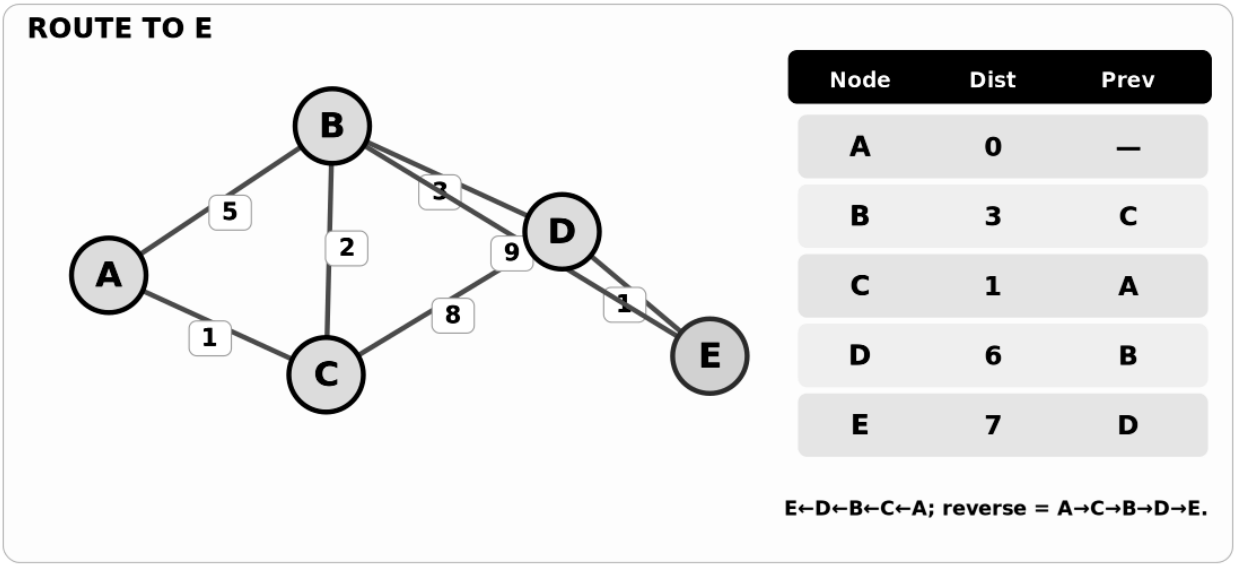


SETTLE D Dijkstra: E improves to 7.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



ROUTE TO E Dijkstra: E←D←B←C←A; reverse = A→C→B→D→E.

I DO — WORKED EXAMPLE

Follow the worked method

Find the shortest route S→T and reconstruct it.

Predict the next state

Start S=0: tentative A=2, B=6.

Update the table

Settle A=2; B improves to 4 and T receives 9.

Update the table

Settle B=4; T improves to 5.

Update the table

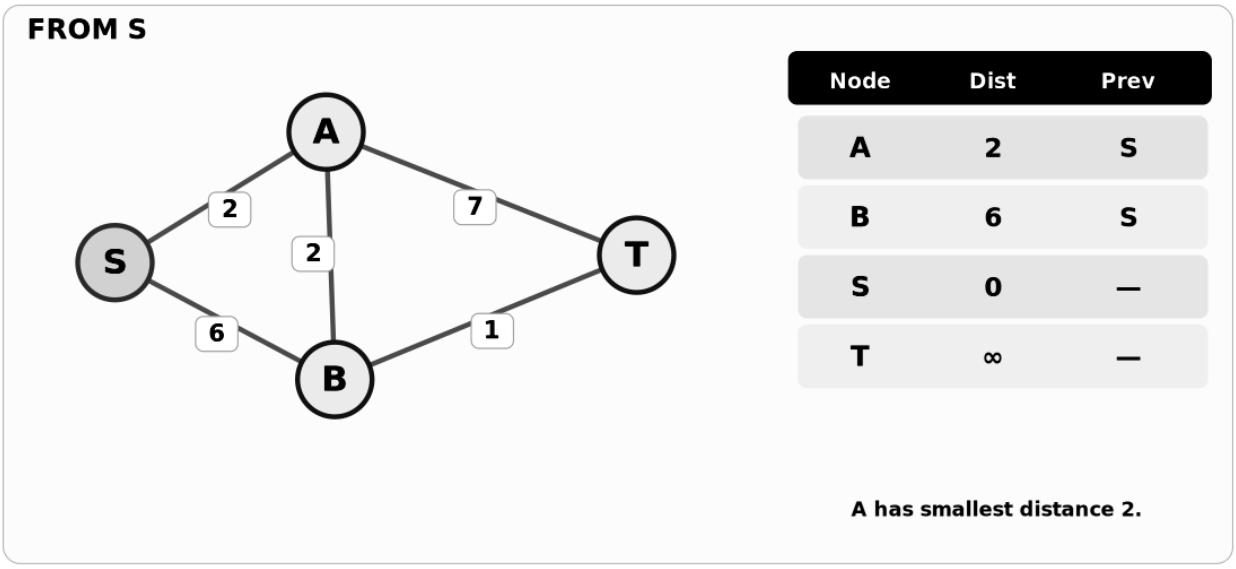
Trace T←B←A←S, then reverse: S→A→B→T at cost 5.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find the shortest route S→T and reconstruct it.

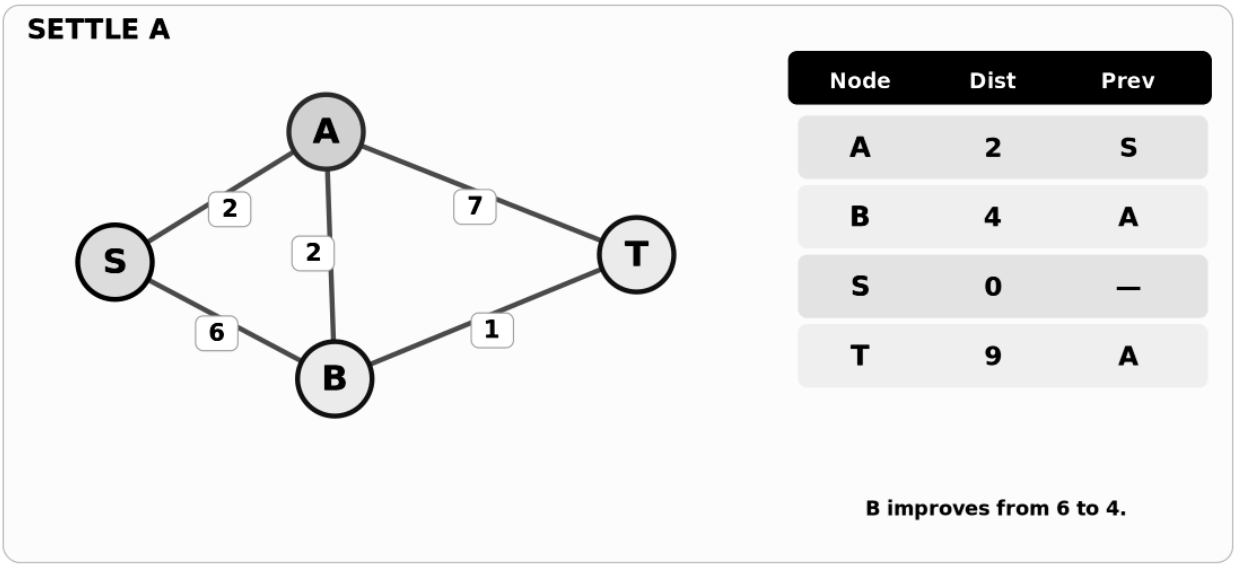
Start S=0: tentative A=2, B=6.



FROM S Dijkstra: A has smallest distance 2.

Find the shortest route S→T and reconstruct it.

Start S=0: tentative A=2, B=6.



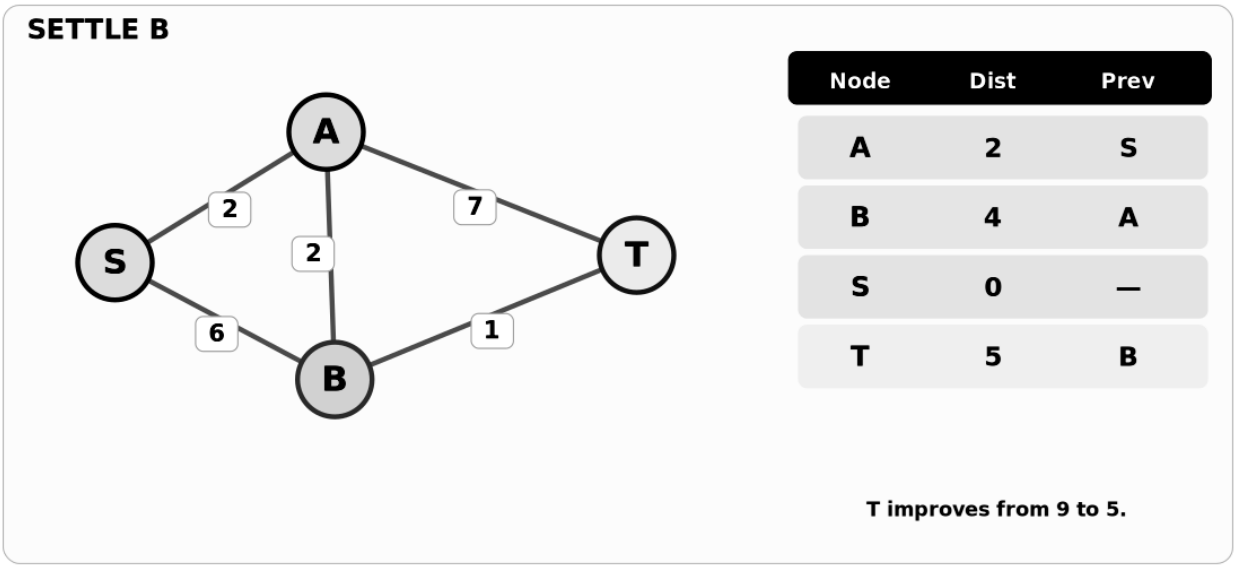
SETTLE A Dijkstra: B improves from 6 to 4.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find the shortest route S→T and reconstruct it.

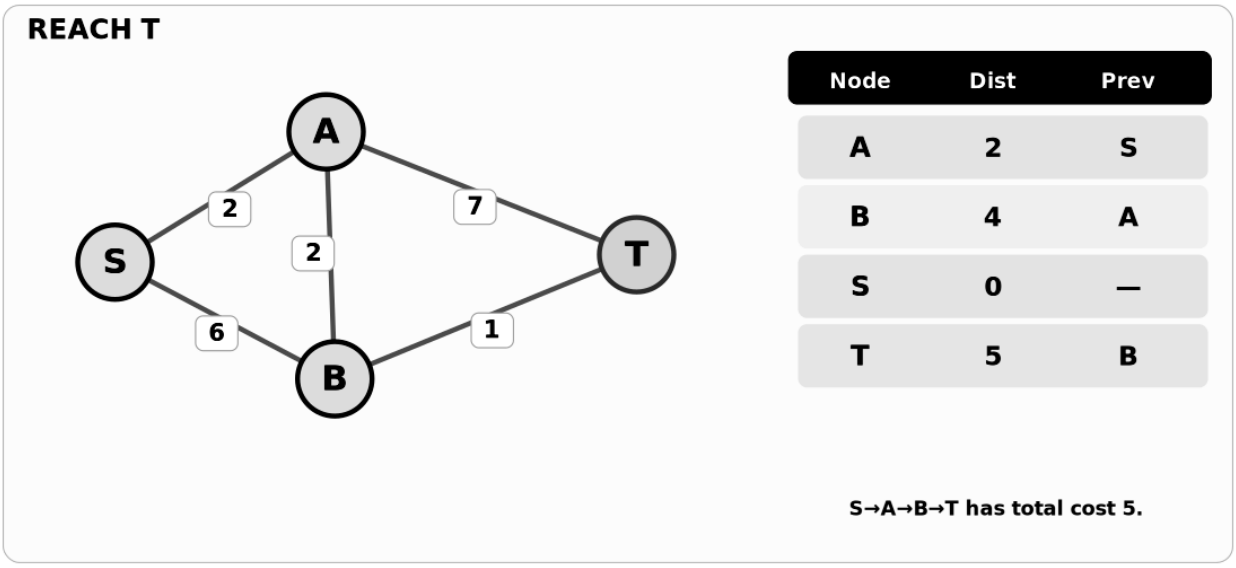
Start S=0: tentative A=2, B=6.



SETTLE B Dijkstra: T improves from 9 to 5.

Find the shortest route S→T and reconstruct it.

Start S=0: tentative A=2, B=6.



REACH T Dijkstra: S→A→B→T has total cost 5.

WE DO — GUIDED PRACTICE

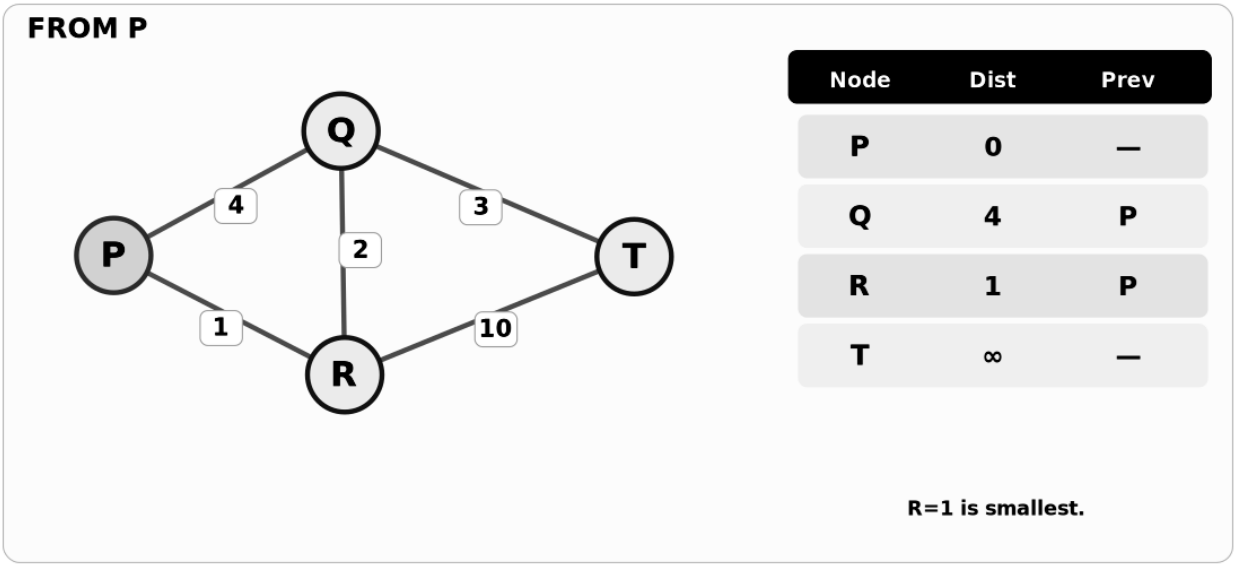
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find the shortest route P→T and reconstruct the predecessors.

Start P=0 gives Q=4, R=1; settle R next.



FROM P Dijkstra: R=1 is smallest.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict the next state Start P=0 gives Q=4, R=1; settle R next.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Starting at U, edges U–V=4, U–W=1 and W–V=2. Find the shortest U→V path and total cost.

Question 2

Edges S–A=5, S–B=1, B–A=2, A–T=3 and B–T=9. Find S→T shortest distance, route and predecessor of T.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A graph contains a disconnected vertex Z. Explain what happens to tentative distance and predecessor for Z in Dijkstra, and when the algorithm may stop.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A predecessor table should be followed from the destination backwards before reversing.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

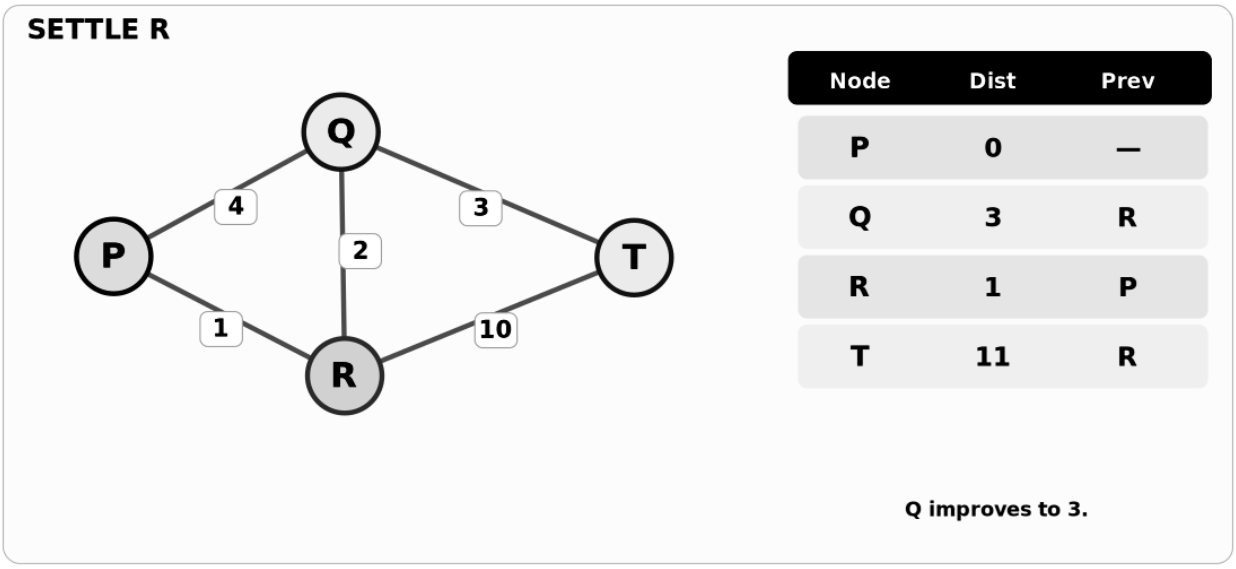
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find the shortest route P→T and reconstruct the predecessors.

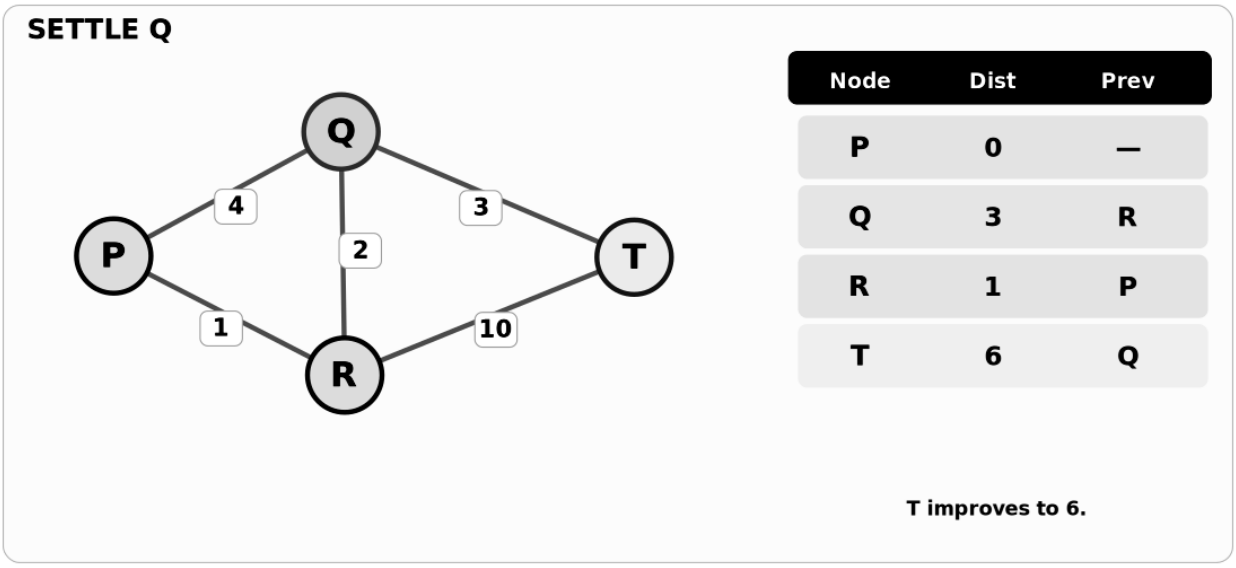
Start P=0 gives Q=4, R=1; settle R next.



SETTLE R Dijkstra: Q improves to 3.

Find the shortest route P→T and reconstruct the predecessors.

Start P=0 gives Q=4, R=1; settle R next.



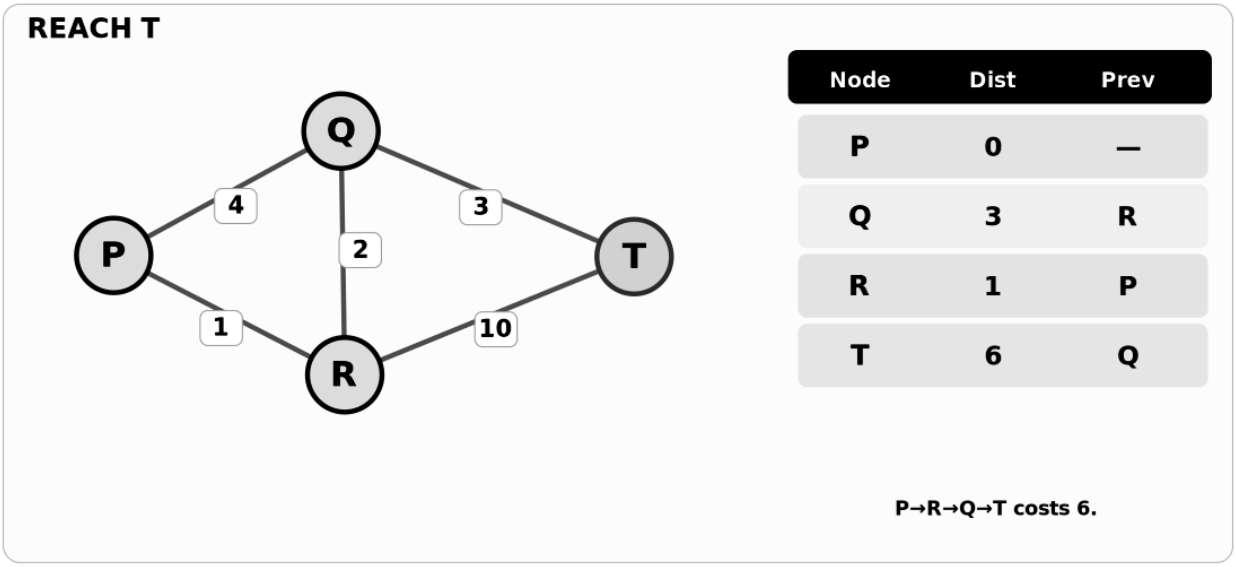
SETTLE Q Dijkstra: T improves to 6.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find the shortest route P→T and reconstruct the predecessors.

Start P=0 gives Q=4, R=1; settle R next.



REACH T Dijkstra: P→R→Q→T costs 6.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Update the table

R improves Q to 3 and gives T=11.

Q improves T to 6.

T←Q←R←P reverses to P→R→Q→T, cost 6.

IMPROVE: Explain one correction to your own reasoning.